

CASE STUDY

Predictive Maintenance — FMCG Bottling Plant

Case Study | Chennai, Tamil Nadu | 2023

Executive Summary

A leading FMCG company's high-speed PET bottling facility in Chennai was suffering from frequent unplanned stoppages on its filling and capping lines, eroding throughput and causing costly emergency maintenance callouts. Our predictive maintenance solution—combining vibration sensors, thermal imaging, and ML-based failure prediction—reduced unplanned downtime by 40% and improved overall productivity by 12% within nine months of deployment.

40%	12%	<3%	9 mo
Unplanned Downtime ↓	Productivity Gain	False Alarm Rate	Deployment

Client Context

The Chennai facility runs six high-speed bottling lines at 36,000 bottles per hour each. Line stoppages during peak production windows (festival seasons) directly impacted customer service levels. The maintenance team relied on fixed-schedule preventive maintenance, resulting in unnecessary part replacements and missed early failure signals.

Scope of Work

- Deployment of 380 vibration sensors and 60 thermal imaging cameras across 6 bottling lines
- Edge computing nodes (NVIDIA Jetson) for real-time signal processing at line level
- ML failure-prediction models trained on 18 months of historical breakdown records
- Automated work-order generation integrated with SAP PM upon anomaly detection
- Maintenance crew mobile app for alert acknowledgment and root-cause capture
- Digital twin of Line 2 for maintenance simulation and spare-part optimisation

South India Delivery Context

This engagement was delivered from our Chennai centre of excellence, which hosts a dedicated South India manufacturing practice of 45+ consultants. Local delivery enabled rapid on-site iteration and deep collaboration with the client's Tamil Nadu-based engineering team, reducing response time for critical issues to under 2 hours.

Key Outcomes

- Unplanned downtime reduced by 40% across all 6 bottling lines
- 12% productivity improvement translating to additional 2.1 million bottles per month
- False alarm rate maintained below 3%, preserving maintenance crew trust in the system
- Spare parts inventory cost reduced by █1.4 Cr annually through condition-based ordering

- System expanded to 2 additional plants in Tamil Nadu within 6 months of initial go-live